복사전달 특강
Special Topics in Radiative Transfer

Lecture 8
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Based on: Chapters 5 & 6 of Noebauer & Sim (Living Reviews in Computational Astrophysics)

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Chapter 5: Monte Carlo Quanta

Overview of discretization paradigms

What are MC Quanta?

- MCRT discretizes radiation into a large number of packets.
- Each packet carries: position, propagation direction, frequency, energy/weight, and (optionally) a Stokes polarization vector.

5.1 Photon Packet Scheme

- Quanta = bundles of physical photons
- Weights encode photon count per packet
- Non-uniform & variable weights → variance reduction
- Dominant in dust RT simulations

5.2 Energy Packet Scheme

- Quanta = parcels of radiant energy
- Packet energy = its weight (conserved)
- Indivisible: no splitting during propagation
- Developed by L. Lucy (1999)

Key difference: photon packets conserve photon number; energy packets conserve energy.

5.1 Photon Packet Scheme

Historically dominant approach in astrophysical MCRT

Historical Inspiration

- Inspired by nature's own discretization.
- Early MCRT studies (Auer 1968; Avery & House 1968) referred to quanta simply as 'photons'.

Monte Carlo photon packets

- Each MC packet represents a vast number of real photons.
- The simulation count is completely insignificant compared to the actual radiation field.

Packet Weights

- Weights encode the physical photon count per packet.
- In this scheme, the MC quanta are typically referred to as ***photon packets*** or simply ***packets***. Non-uniform and variable weights reduce variance (MC noise) dramatically — a form of biasing (Sect. 9.4).

Applications

- Widely used in dust RT (Whitney 2011; Steinacker et al. 2013).
- Most astrophysical MCRT codes rely on this scheme with sophisticated weight manipulations.

5.2 Energy Packet Scheme

Lucy's indivisible energy packet formalism

Core Principle

- Packets are parcels of radiant energy.
- The packet's energy serves as its weight and is strictly conserved — packets are **indivisible** and **indestructible** (except at domain boundaries).

Indivisibility

- No splitting.
- Multiple outcome channels handled by probabilistic branching, preserving total energy throughout.

Energy Conservation

- Strict local energy conservation by construction in RE problems.
- Frequency can change at each interaction; CMF energy stays fixed.

Photon Number

- NOT conserved (physically correct: e.g. recombination cascades don't conserve photon number).
- A packet may represent varying photon counts during its lifetime.

- Many physical radiation-matter processes (e.g. recombination cascades or fluorescence) do not conserve the number of photons.

Key Advantages

- Effort proportional to energy carried, not photon count
- Avoids accumulation of negligible-weight quanta (no elimination scheme needed)
- Widely used: stellar winds, SN ejecta (Lucy 2005; Kerzendorf & Sim 2014)

Compton Scattering: Two Schemes Compared

A worked example illustrating the philosophical difference

Process: $e_i^- + \gamma_i \;\rightarrow\; e_f^- + \gamma_f$ *(photon transfers recoil energy to electron;* $\nu_f < \nu_i$*)*

Photon Packet Approach

- Packet scatters → photon number fixed
- Frequency reduced ($\nu_f < \nu_i$)
- Packet energy decreases accordingly

✓ Good for: computing Comptonized spectra

✗ Risk: many low-energy packets accumulate, each costing the same compute per scatter

Energy Packet Approach

- Track energy flow, not photon count
- $F_\gamma = E_f^\gamma / E_i^\gamma$ of the incident photon energy → is passed to the outgoing photon packet.
- $F_e = E_f^e / E_i^\gamma = 1 - F_\gamma$ → goes to the electron kinetic pool (*k*-packet)

✓ Energy strictly conserved

✓ Effort ∝ energy carried (ideal for heating applications)

Which is better?

- Neither is universally superior. Photon packets suit spectral calculations (Pozdnyakov et al. 1983).
- Energy packets suit heating applications (SN ejecta powered by radioactive decay, Lucy 2005).

5.3 Packet Initialization

Assigning initial properties via Monte Carlo sampling

Each packet is initialized with:

Position · Propagation direction · Frequency · Energy / Weight · Stokes vector (if polarization included)

Case A: Uniform, Thermal Radiation from a volume ΔV

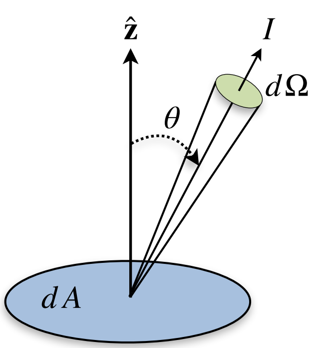
- Each of N packets carries: $\varepsilon = \Delta V \cdot a_R T^4 / N$
  $(a_R = (4/c)\sigma_{SB})$
- Position: uniform random in cuboid ΔV
  $x = x_0 + \xi_1(x_1 - x_0)$
  $y = y_0 + \xi_2(y_1 - y_0)$
  $z = z_0 + \xi_3(z_1 - z_0)$
- Direction: isotropic (Eq. 26-27): $\mu = 2\xi_1 - 1$ and $\phi = 2\pi\xi_2$
- Frequency: sampled from Planck function

Case B: Photospheric Boundary

- $N = 4\pi R^2_{\mathrm{phot}} \sigma_{\mathrm{SB}} T^4_{\mathrm{phot}} \Delta t / \varepsilon$
- Position: $r = R_{\mathrm{phot}}$ (on the photosphere)
- Direction: $\mu = \sqrt{\xi_1}$ and $\phi = 2\pi\xi_2$
  (flux-based; limb darkening neglected; isotropic radiation)
- Frequency: sampled from Planck function

- *Initialization is performed in the co-moving frame (CMF).*
- *Lab-frame (LF) properties are obtained via frame transformations (see Sect. 8 for expanding media).*

$\mu = \cos\theta$



Contribution of I to the flux:
$P(\mu,\phi)d\mu d\phi dA = I \cdot d\mu d\phi (dA \cdot \mu)$
for $0 \le \mu \le 1$ $(0^\circ \le \theta \le 90^\circ)$

$\therefore P(\mu,\phi) = \mu \dfrac{1}{2\pi}$ after normalization

$\int_0^{\mu} d\mu' \mu' = \xi_1$ and $\int_0^{\phi} d\phi' = \xi_2$

Chapter 6: Propagation of Quanta

The core computational loop of any MCRT simulation

- After initialization, packets are propagated through the domain.
- This constitutes the bulk of MCRT runtime.

1. Draw τ

Sample optical depth:

$$\tau = -\ln \xi$$

2. Locate interaction

Invert $\int_0^l dl' \chi(l') = \tau$

to find distance l.

Here, $\chi(l')$ is the opacity.

3. Compare distances

distance l vs. cell boundary,
vs. time step end
→ take minimum,
assign the shortest distance to l.

4. Move packet

- Advance by l
- Update time stamp $t = t + l/c$

5. Perform event

- Scattering
- absorption, or
- cell crossing

6. Termination?

- Escape through one of the domain boundaries
- Absorption interaction
- Reaching the end of a pre-defined time interval.

Fig. 2: Packet Propagation Algorithm

The two-loop structure of the MCRT propagation algorithm

Outer Loop: Packet Loop

- Iterates over all MC packets in the population
- Each packet processed through propagation loop
- Returns to outer loop once packet terminates

Inner Loop: Propagation Loop

- Moves packet step by step through domain
- Each step: draw τ → invert → move → event
- Loop continues until termination condition

Update Estimators

- At each step, packet properties are recorded to reconstruct radiation field quantities (energy density, flux, etc.) from the ensemble. See Sect. 9.

Monte Carlo radiative transfer

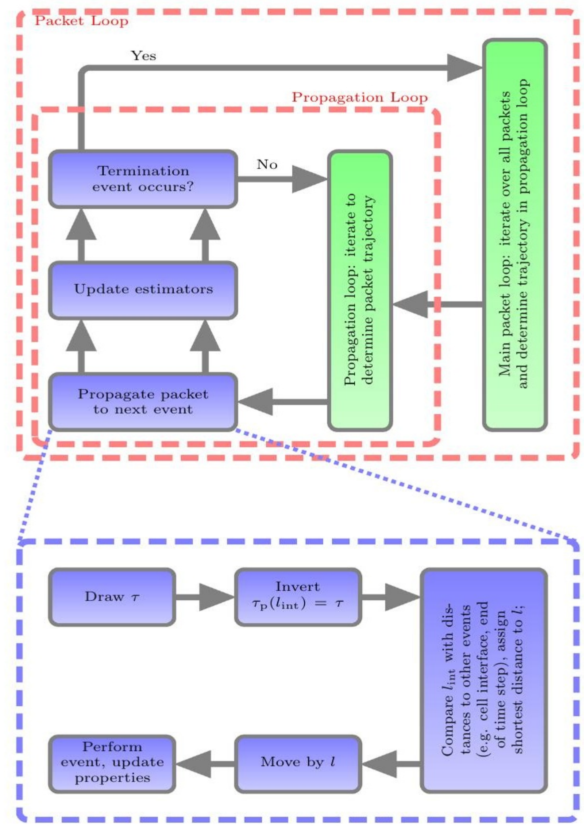


Fig. 2 Illustration of the packet propagation process. At its core, the main processing loop works through the entire packet population, propagating each packet individually. In this loop, each packet is moved through the domain until a termination event occurs. Depending on the specific MCRT application, this may for example be an absorption interaction, escaping through one of the domain boundaries or reaching the end of a pre-defined time interval. The instruction "update estimators" refers to all activities related to recording packet properties that are used to reconstruct physical information about the radiation field from the packet ensemble. This procedure will be discussed in more detail in Sect. 9

6.2 Absorption: Discrete vs. Continuous

Two complementary approaches to handling absorptive opacity

Discrete Interaction Events

- Packet propagates until τ threshold reached
- Absorption terminates packet at that point
- In RE (Lucy 1999a): immediately re-emitted
- Required for line interactions where absorption ↔ re-emission must be linked
- Used in: Abbott & Lucy 1985; Long & Knigge 2002

Continuous Weight Degradation

- Weight decays along flight path:

  $w(l) = w_0 \cdot \exp(-\tau_a)$, where $\tau_a(l) = \int_0^l dl' \chi_a(l')$ is the optical depth due to the absorption opacity χ_a.
- Smoother → lower MC noise for small opacity
- Disconnects absorption from re-emission; new packets must be injected separately
- Suits pure-absorption problems
- Hybrid schemes common (continuum + lines)

This review focuses primarily on the discrete interaction approach; most principles also apply when weight degradation is used for continuum opacity.

6.3 Numerical Discretization & Material Properties

Implementing MCRT on a computational grid

- Real applications require a spatial grid dividing the domain into cells. Material properties (density, temperature, velocity) are represented discretely per cell.

Material State per Cell

- Sets local opacity → rate of optical depth accumulation [Eq. 44].
- Also dictates re-emission characteristics in scattering events.

Cell Crossing Events

- When a packet crosses a cell boundary, opacities are recalculated.
- τ may be redrawn (exploiting infinite divisibility of the exponential distribution).

Opacity Integration

- Within a constant-opacity cell: $l = \tau/\kappa$ (trivial inversion of Eq. 44).
- More complex for fast flows where opacity is frequency/direction dependent (Sect. 8).

Grid Types

- Cartesian, spherical, cylindrical, or unstructured — chosen based on problem geometry.
- Interpolation between cells is possible but adds computational cost.

6.4 Absorption & Scattering Interactions

Handling multiple opacity sources probabilistically

- Total opacity: $\kappa_{\mathrm{tot}} = \kappa_{\mathrm{s}} + \kappa_{\mathrm{a}}$
- Interaction distance: $l = \tau / \kappa_{\mathrm{tot}}$

Decision at the interaction point:

Draw $\xi \in$ [0, 1)

If $\xi \leq \kappa_{\mathrm{s}}/(\kappa_{\mathrm{s}} + \kappa_{\mathrm{a}})$ → SCATTER

Else → ABSORB (remove packet, or re-emit in RE applications)

After Scattering (isotropic coherent example):

- New direction drawn: $\mu = 2\xi_1 - 1, \;\; \phi = 2\pi\xi_2$
- Frequency unchanged (coherent); new τ drawn; accumulated optical depth reset
- Multiple scatterings handled naturally — a major strength of MCRT over deterministic methods!
- Easily extended: anisotropic scattering simply changes the directional sampling rule

6.5 Propagation Example: Homogeneous Sphere

MCRT test case — escape probability from a uniform sphere

- Test problem: photon escape from a sphere of uniform density and emissivity.
- Analytic solution available for pure absorption (see Appendix A.1).
- Packets initialized at $r = R\xi^{1/3}$, direction $\mu = 2\xi - 1$.

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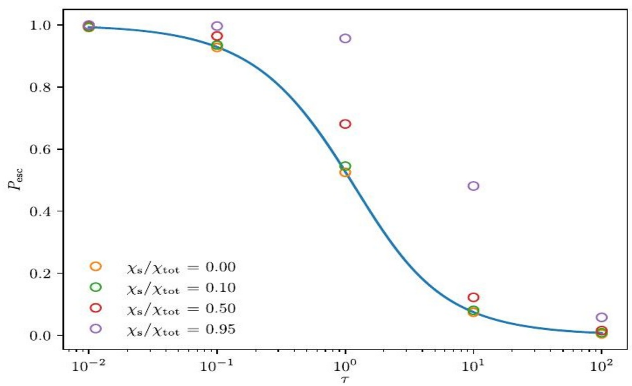

Fig. 3 Results of a number of MCRT simulations for the homogeneous sphere test problem. The escape probability is shown as a function of total optical depth of the sphere, cf. Eq. (55), for different scattering to absorption strengths. For comparison, the analytic solution to the problem with pure absorption is shown as a solid line

Fig. 3: Escape probability $P_{\rm esc}$ vs. total optical depth τ for different scattering fractions. Solid line = analytic solution (pure absorption). MCRT agrees well.

Monte Carlo radiative transfer

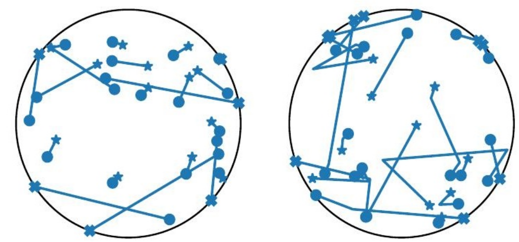

Fig. 4 Illustration of the packet propagation process in the MCRT simulations for the homogeneous sphere problem shown in Fig. 3. A small subset of packet trajectories is visualized for the case $\chi_s = 0$ (left panel) and $\chi_s = \chi_a$ (right panel). The total optical depth of the sphere is in both cases $\tau = 2$. Starting points for the trajectories are marked by small filled circles. The fate of the packet is either marked by a star symbol (absorption) or by a cross (escape)

Fig. 4: Sample packet trajectories for $\tau = 2$. Left: $\kappa_s = 0$ (pure absorption). Right: $\kappa_s = \kappa_a$. ● = start, ★ = absorbed, × = escaped.

6.6 Time-Dependent MCRT

Extending steady-state MCRT to evolving radiation fields

- Time-dependence requires minimal changes: each packet carries an internal clock t, updated as $\Delta t = l/c$ after each propagation step.

Time Stamp

- Each packet: t initialized from emission process.
- Updated by $\Delta t = l/c$.

Time Grid

- Simulation subdivided into time steps t_n.
- Material state constant per step.

Interruption Rule

- Packet reaching t_{n+1} is paused; state saved.
- Resumes in next cycle.

Initialization in time-dependent problems:

- Initial radiation field packets: t = starting time of current simulation phase
- Source packets (continuous emitter): t drawn uniformly over current time step duration
- Material state updated between time steps (coupling with hydrodynamics → Sect. 11)
- Problems arising from very rapid material state changes: addressed in Sect. 10

MCRT example: python code

MCRT test cases

(1) Escape probability from a uniform sphere
https://github.com/unoebauer/mcrtreview-tools

(2) Thompson scattering by an electron gas at temperature T — Team project

Summary

Ch. 5: Two Discretizations

Photon packets conserve photon number; energy packets conserve energy. Choice depends on scientific goal — neither universally superior.

Ch. 5: Initialization

Packets draw position, direction, and frequency from appropriate distributions (Planck for frequency; isotropic or flux-based for direction).

Ch. 6: Propagation Loop

Draw $\tau = -\ln \xi$ → locate interaction → move to minimum-distance event → scatter or absorb → repeat until termination condition.

Ch. 6: Scattering Strength

Multiple, anisotropic scattering is handled naturally in MCRT — a major advantage over deterministic RT solvers with non-local coupling.